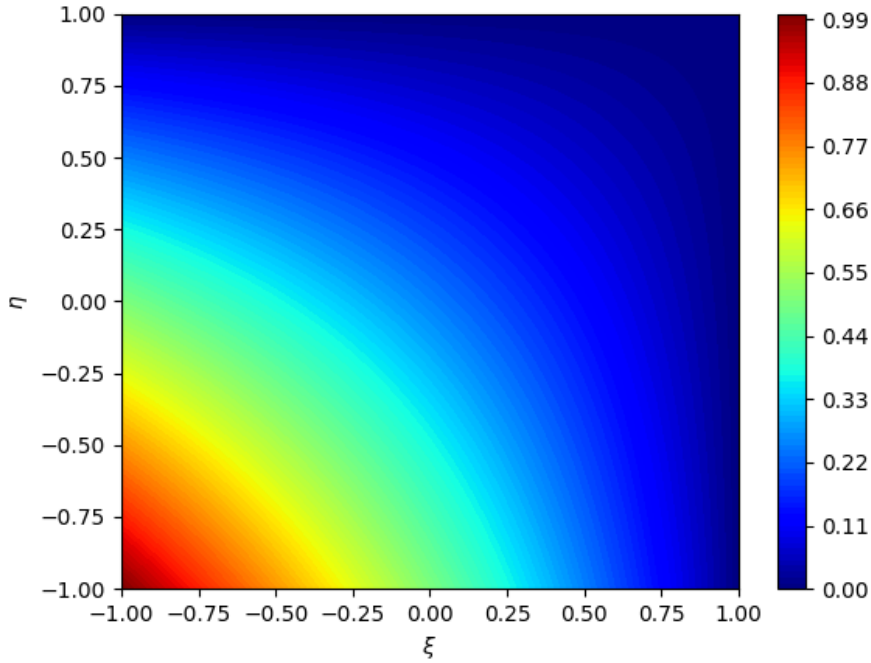
ME EN 507
Christian Valencia Narva

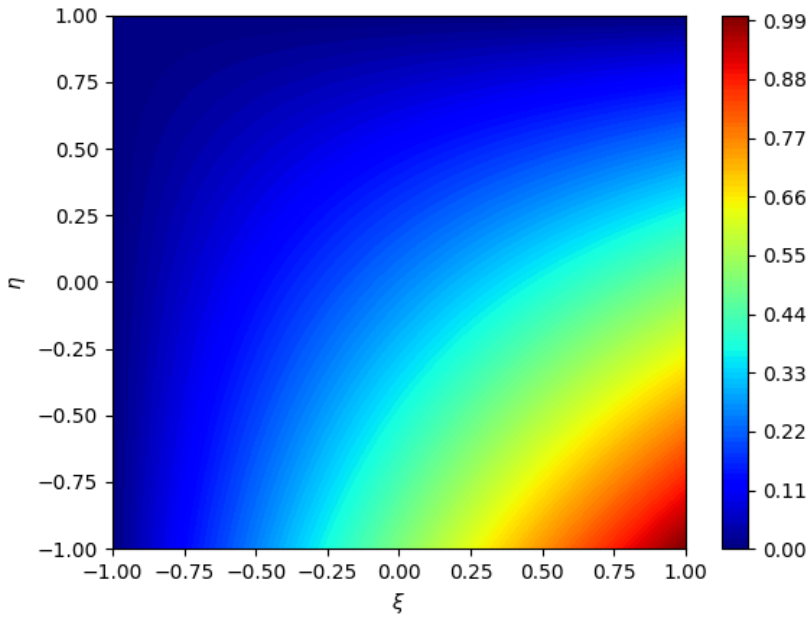
Problem 1

Polynomials of degree [1,1]

A=0



A=1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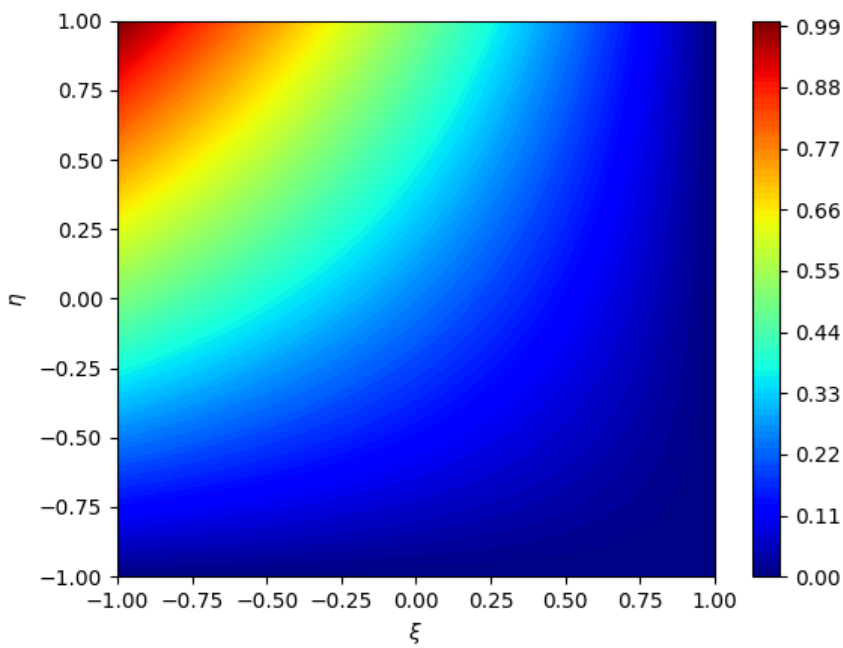
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
η
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
ξ
0.99
0.88
0.77
0.66
0.55
0.44
0.33
0.22
0.11
0.00

A=2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0.75
0.50
0.25
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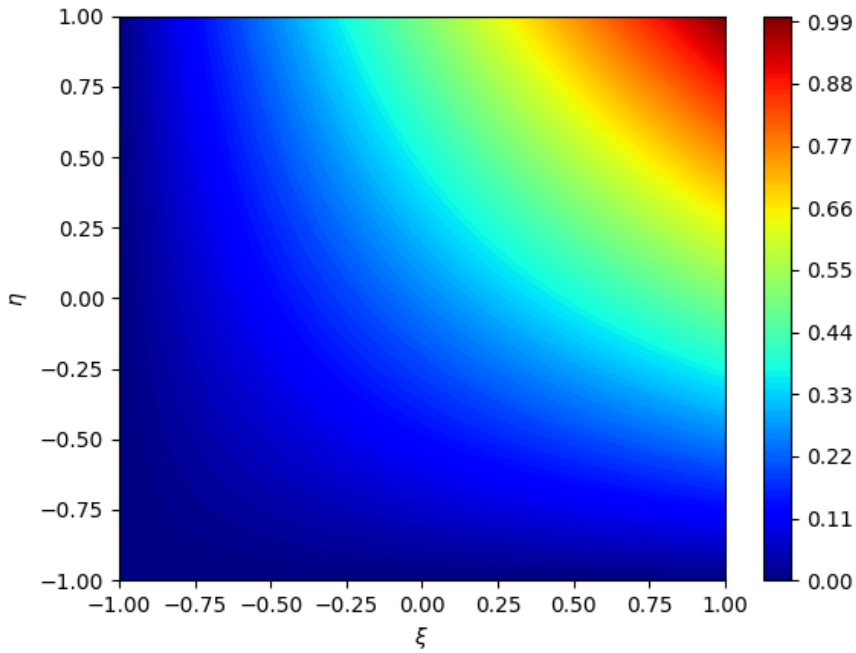
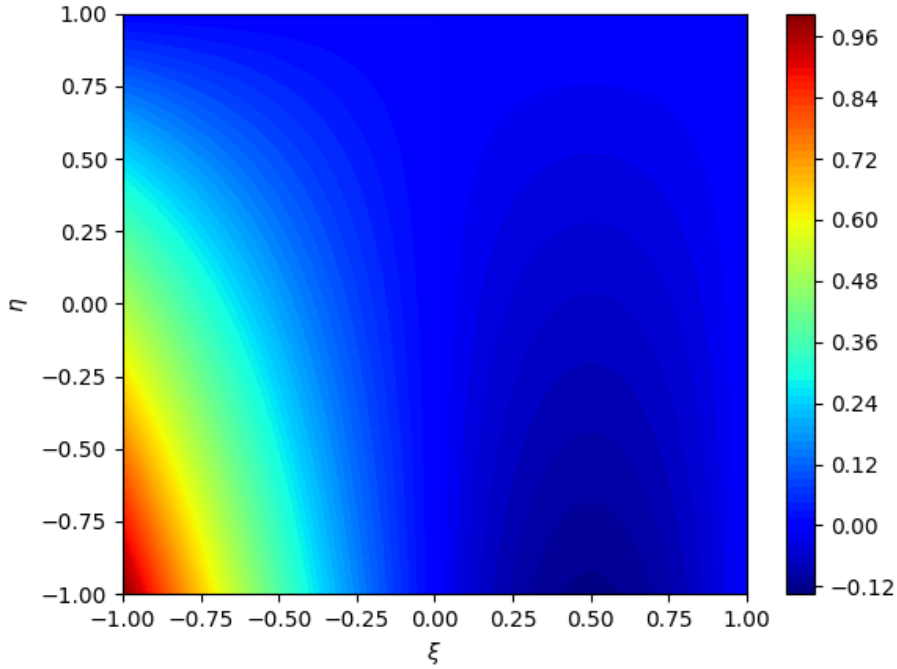
−0.50
−0.75
−1.00
η
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
ξ
0.99
0.88
0.77
0.66
0.55
0.44
0.33
0.22
0.11
0.00

A=3

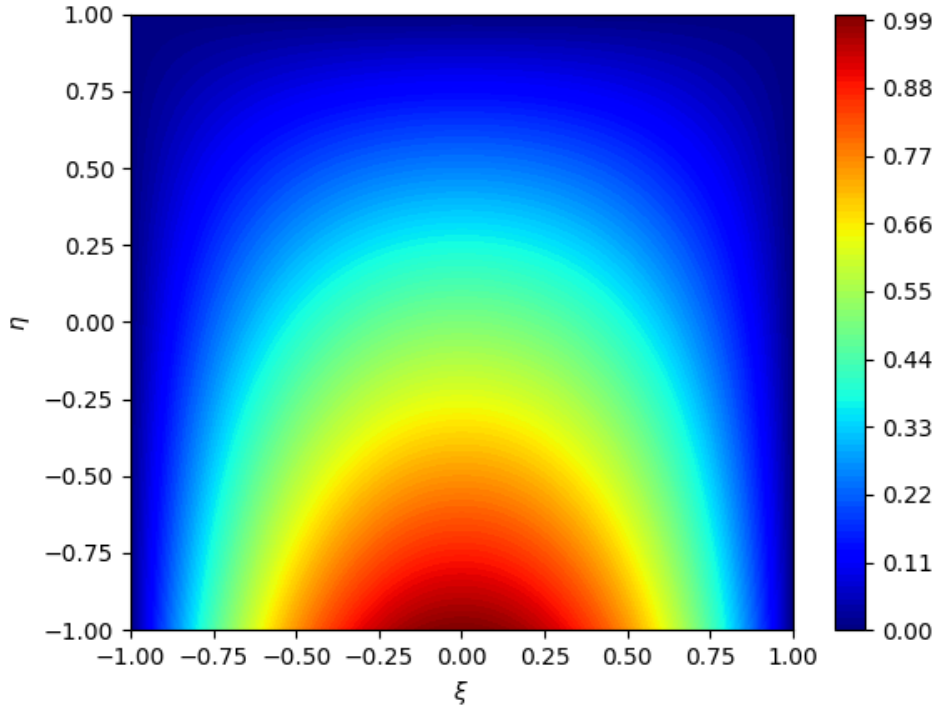


Polynomials of degree [2,1]

A=0



A=1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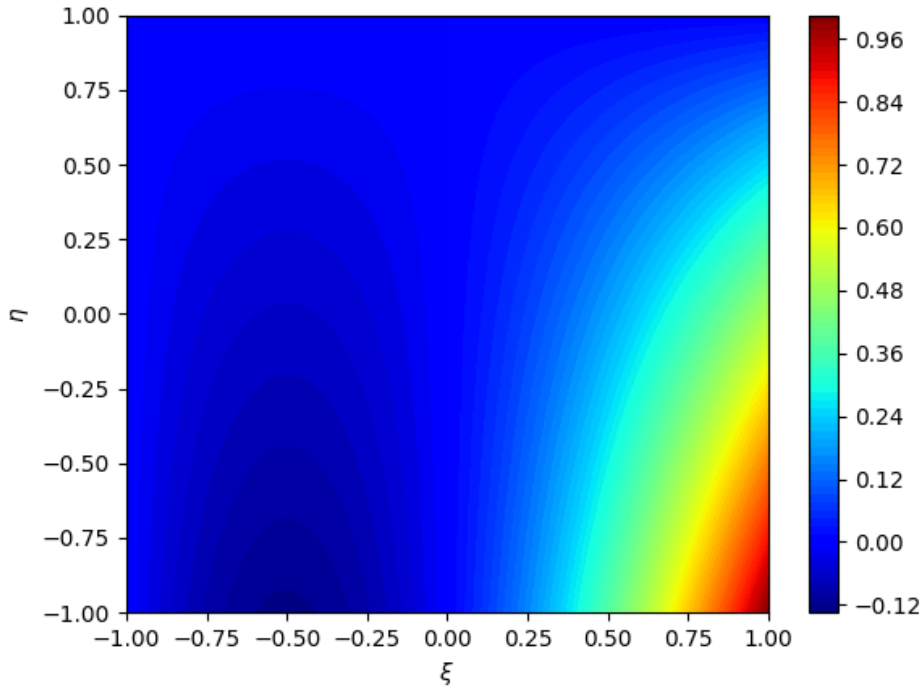
−0.50
−0.75
−1.00
η
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
ξ
0.99
0.88
0.77
0.66
0.55
0.44
0.33
0.22
0.11
0.00

A=2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0.50
0.25
0.00
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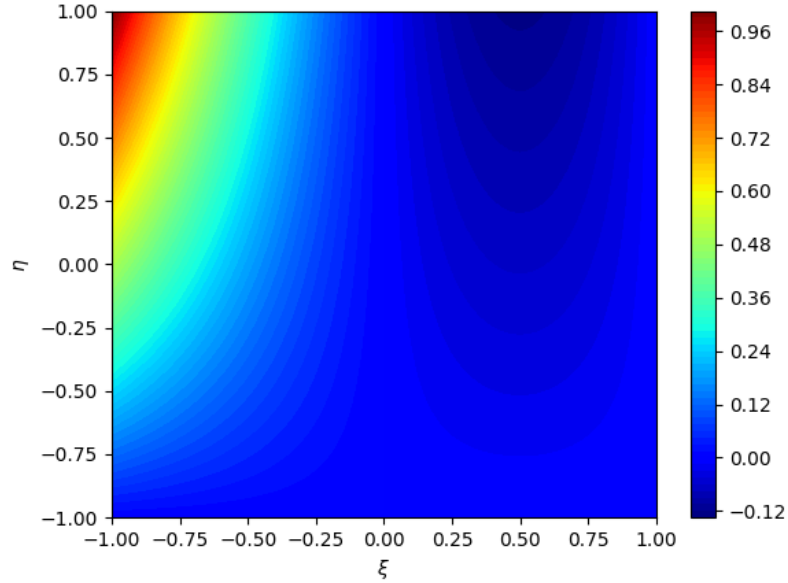
−0.50
−0.75
−1.00
η
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
ξ
0.96
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00
−0.12

A=3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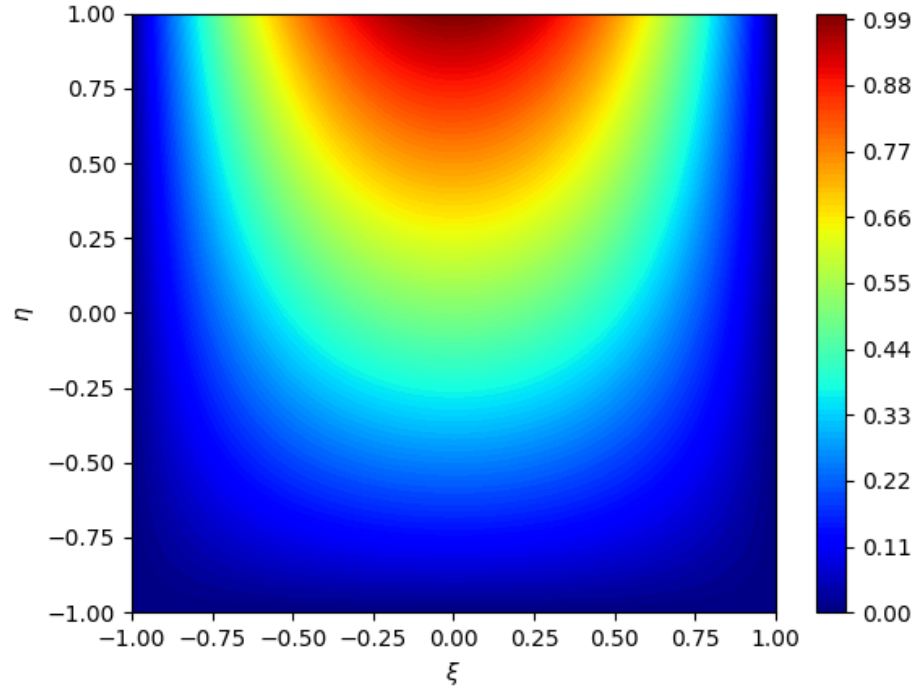
−0.75
−1.00
η
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
ξ
0.96
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00
−0.12

A=4


1.00
0.75
0.50
0.25
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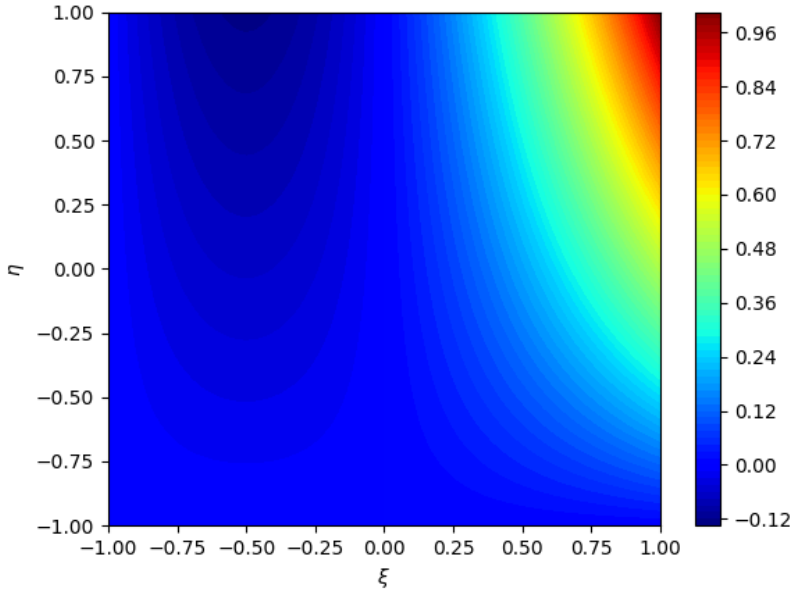
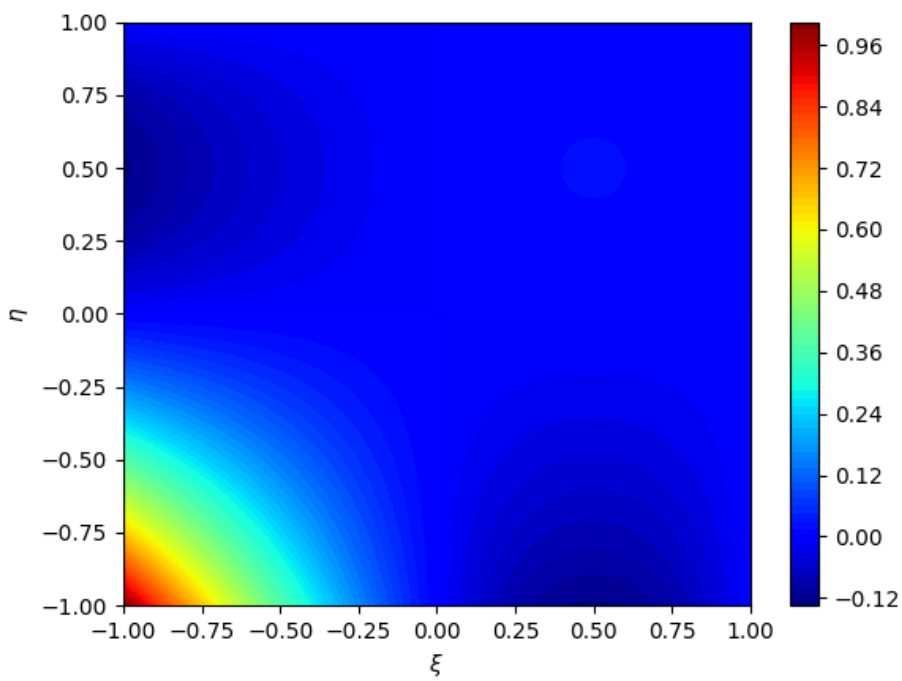
−0.25
−0.50
−0.75
−1.00
η
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
ξ
0.99
0.88
0.77
0.66
0.55
0.44
0.33
0.22
0.11
0.00

A=5

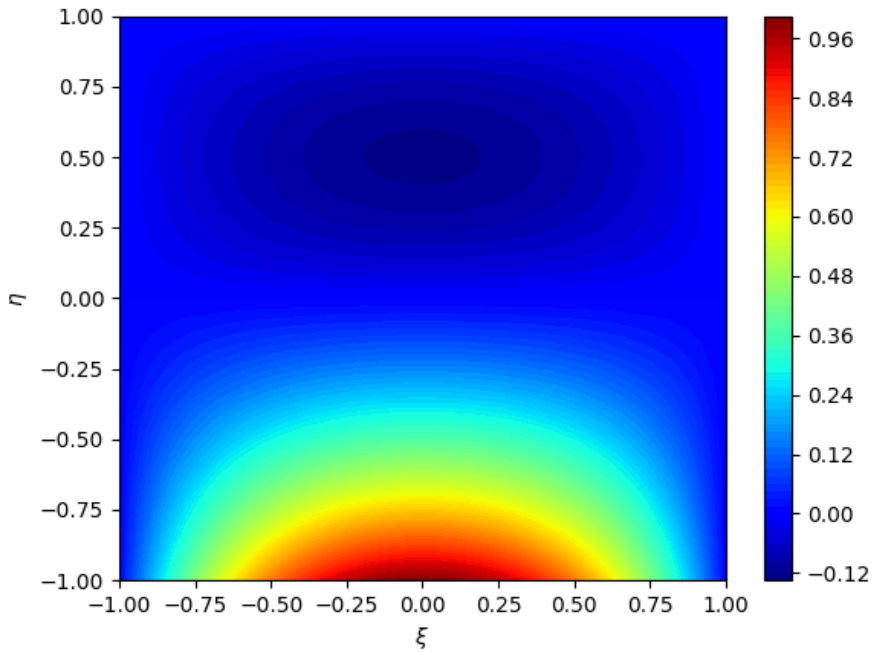


Polynomials of degree [2,2]

A=0



A=1


1.00
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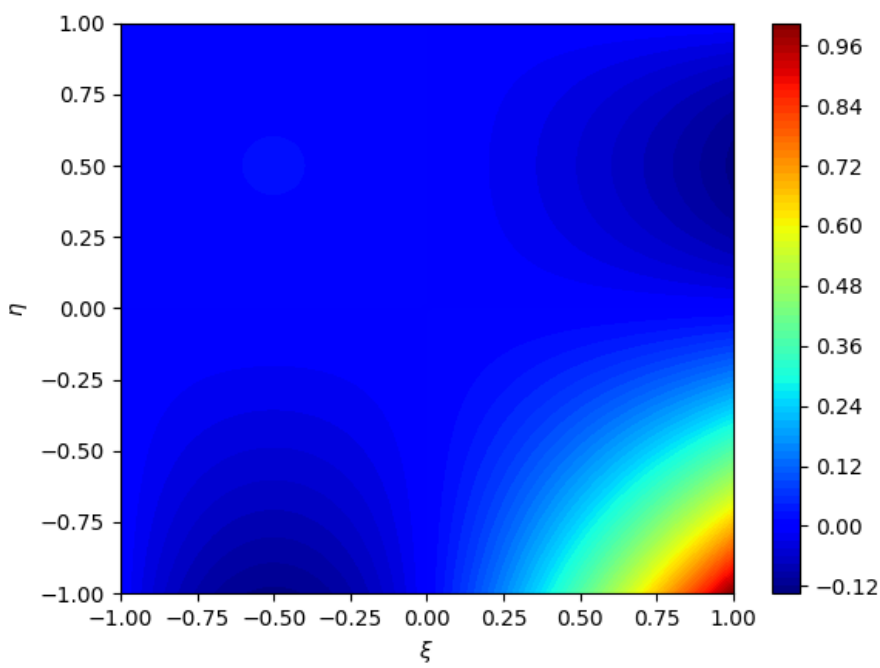
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
η
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
ξ
0.96
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00
−0.12

A=2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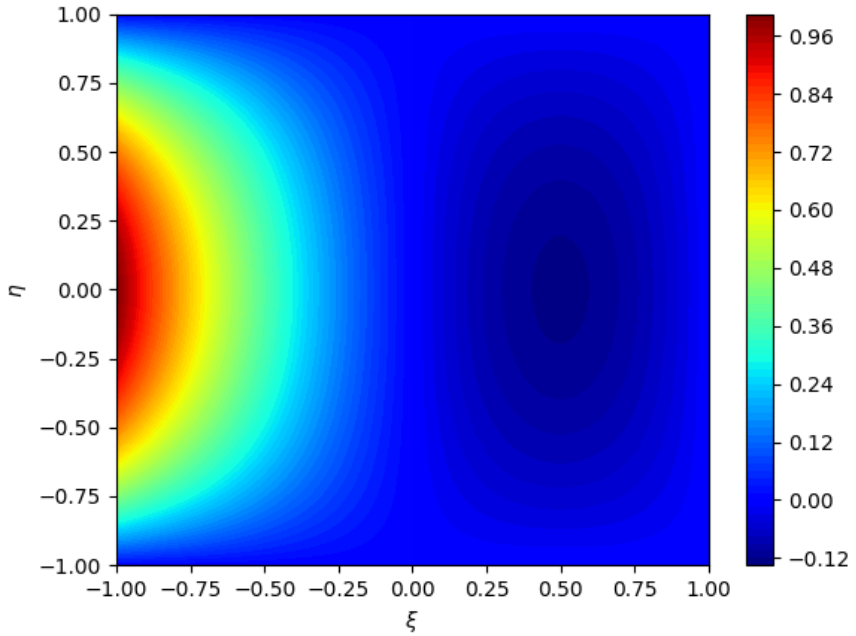
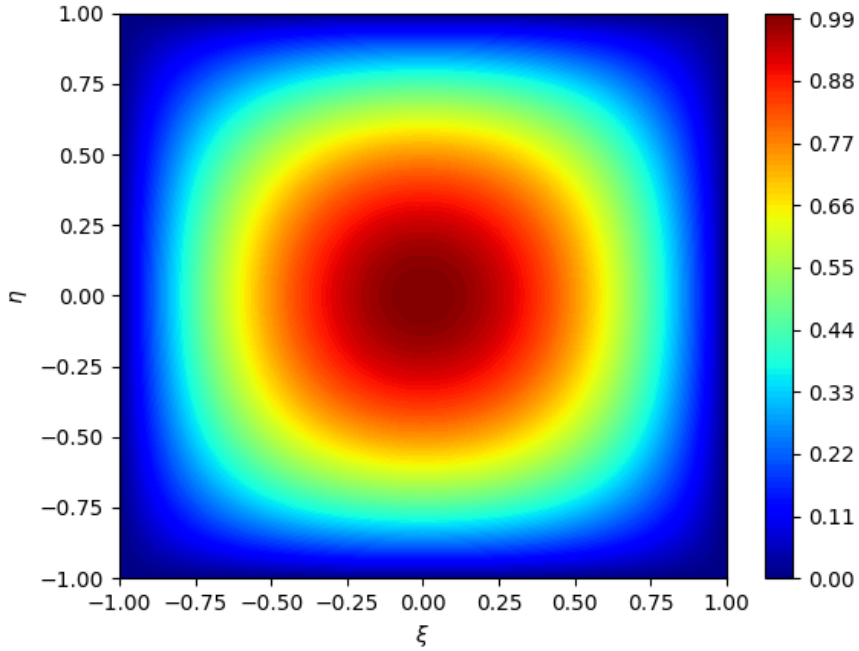
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
η
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
ξ
0.96
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00
−0.12

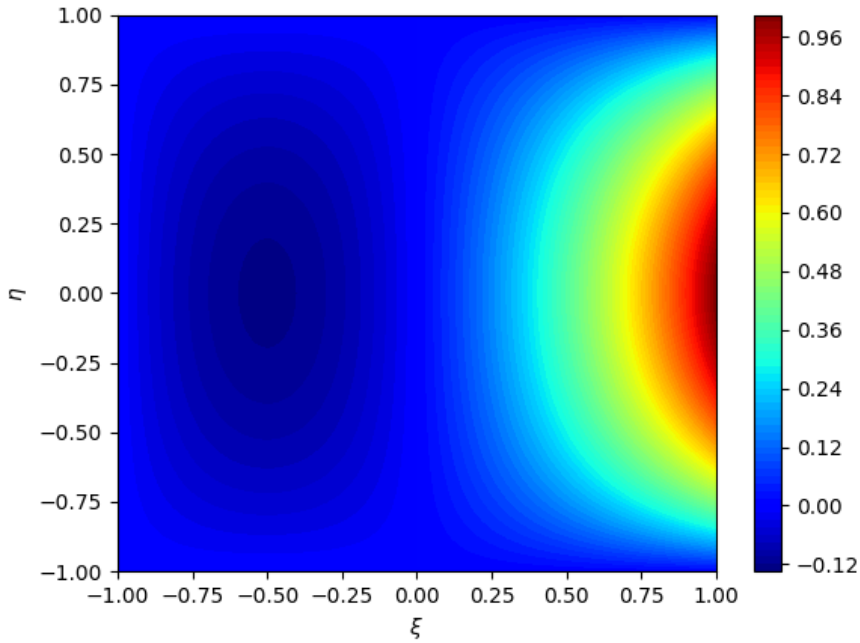
A=3


1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
η
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
ξ
0.96
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00
−0.12

A=4


1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
η
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
ξ
0.99
0.88
0.77
0.66
0.55
0.44
0.33
0.22
0.11
0.00

A=5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0.50
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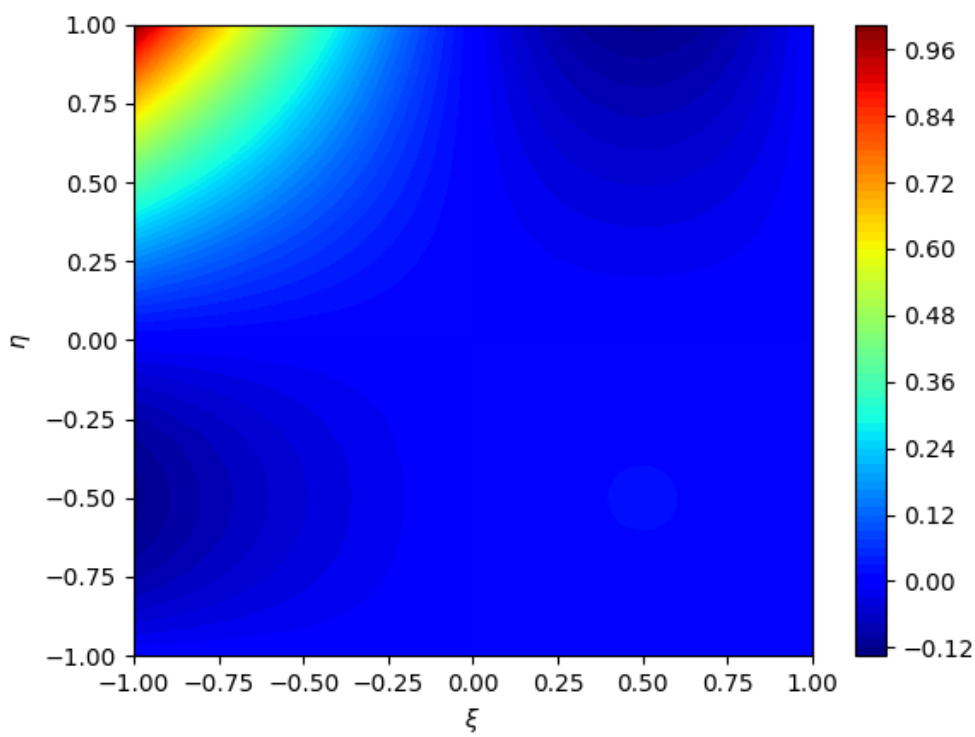
0.00
−0.25
−0.50
−0.75
−1.00
η
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
ξ
0.96
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00
−0.12

A=6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0.50
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0.00
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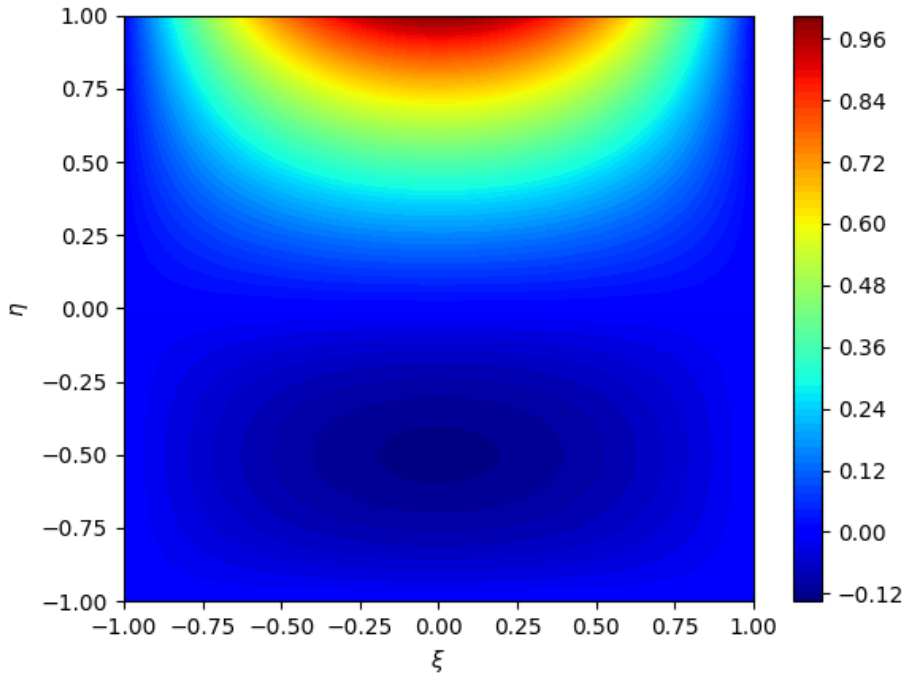
−0.50
−0.75
−1.00
η
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
ξ
0.96
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00
−0.12

A=7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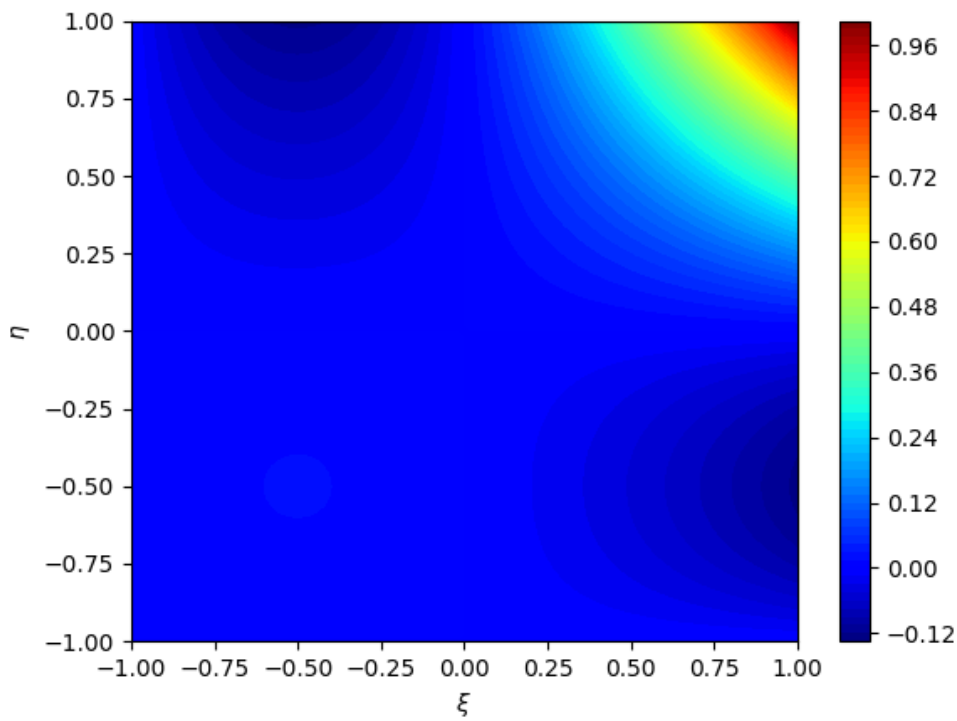
−0.25
−0.50
−0.75
−1.00
η
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
ξ
0.96
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00
−0.12

A=8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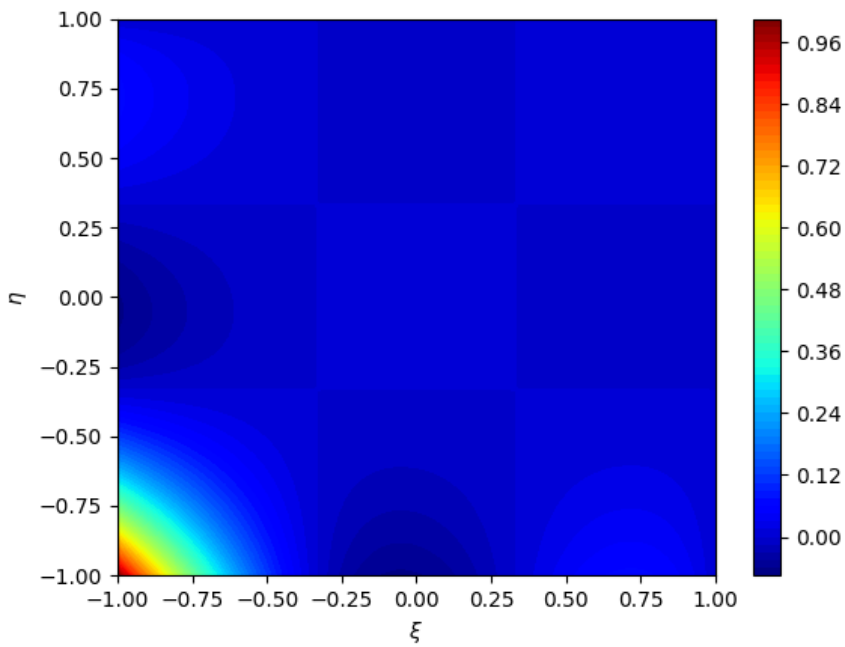
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
η
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
ξ
0.96
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00
−0.12

Polynomials of degree [3,3]

A=0



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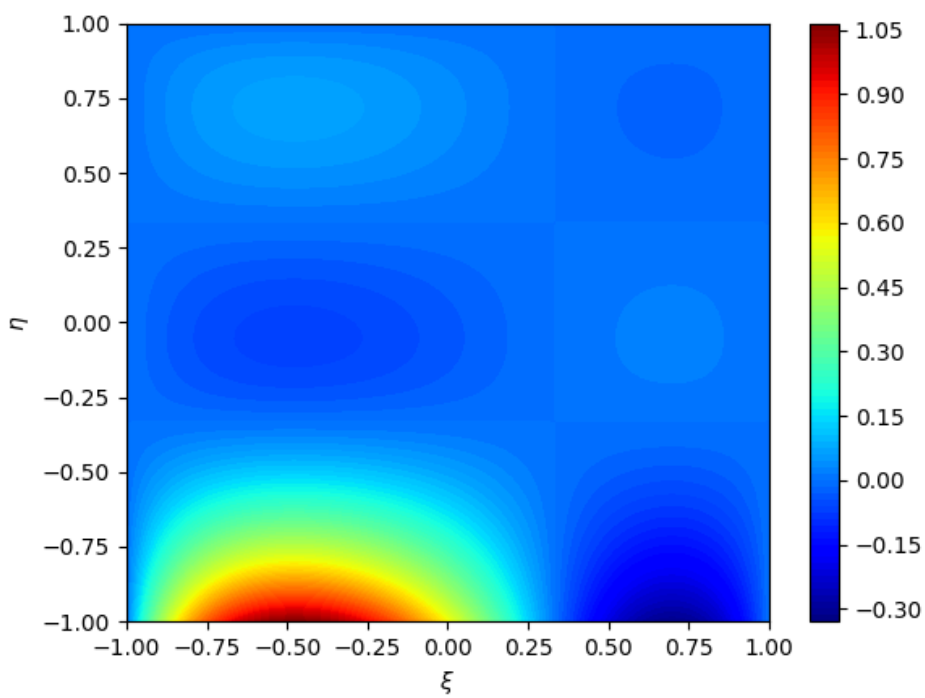

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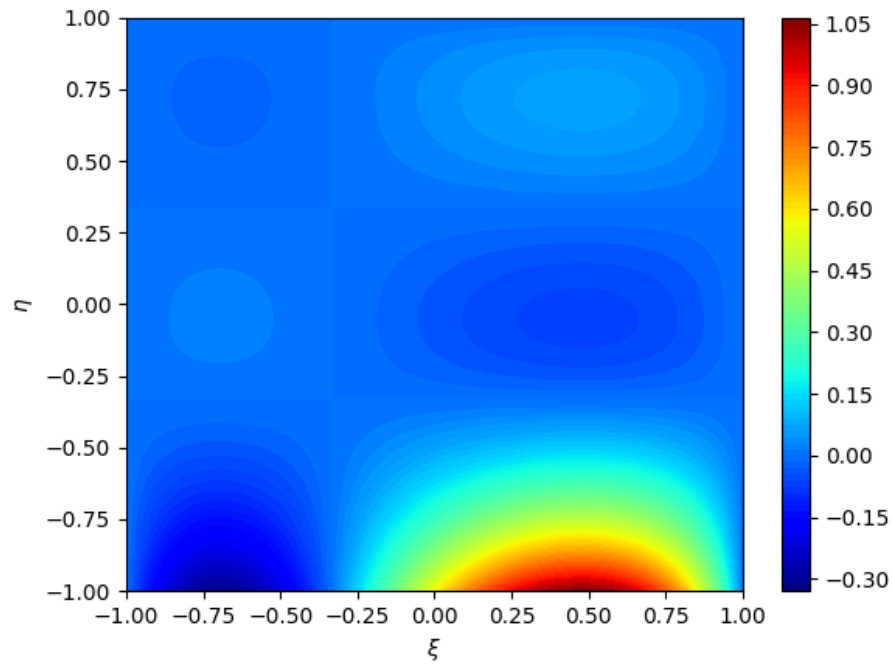
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0.00
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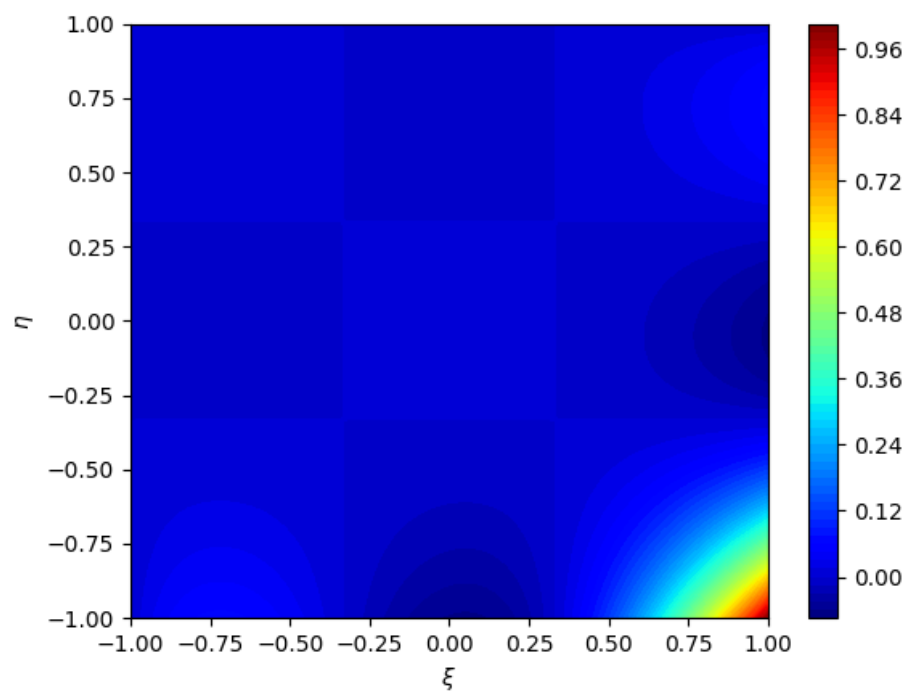
−0.50
−0.75
−1.00
η
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
ξ
1.05
0.90
0.75
0.60
0.45
0.30
0.15
0.00
−0.15
−0.30

A=3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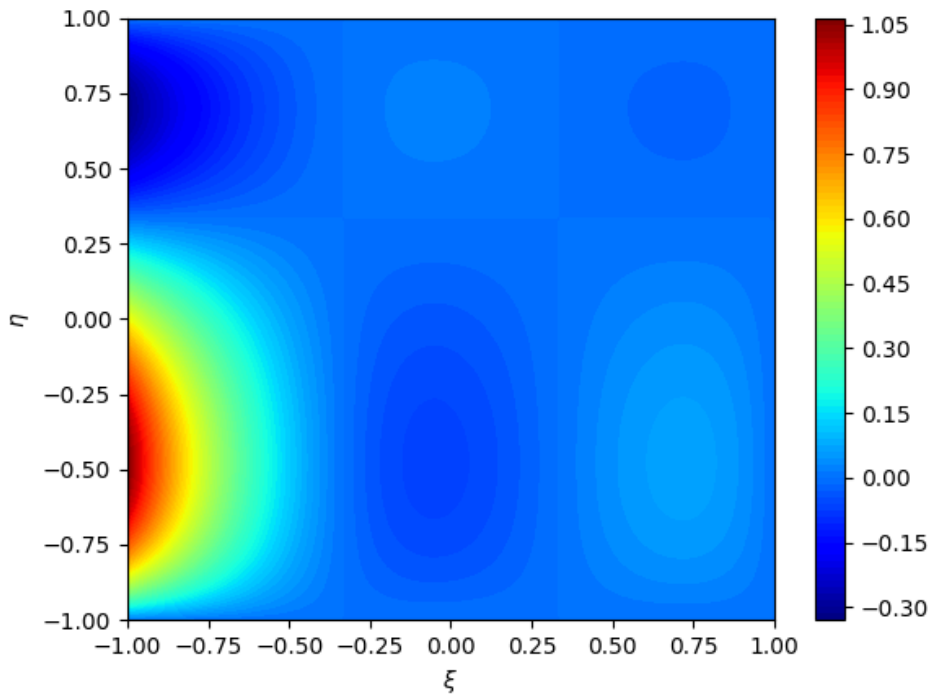
0.25
0.00
−0.25
−0.50
−0.75
−1.00
η
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
ξ
0.96
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00

A=4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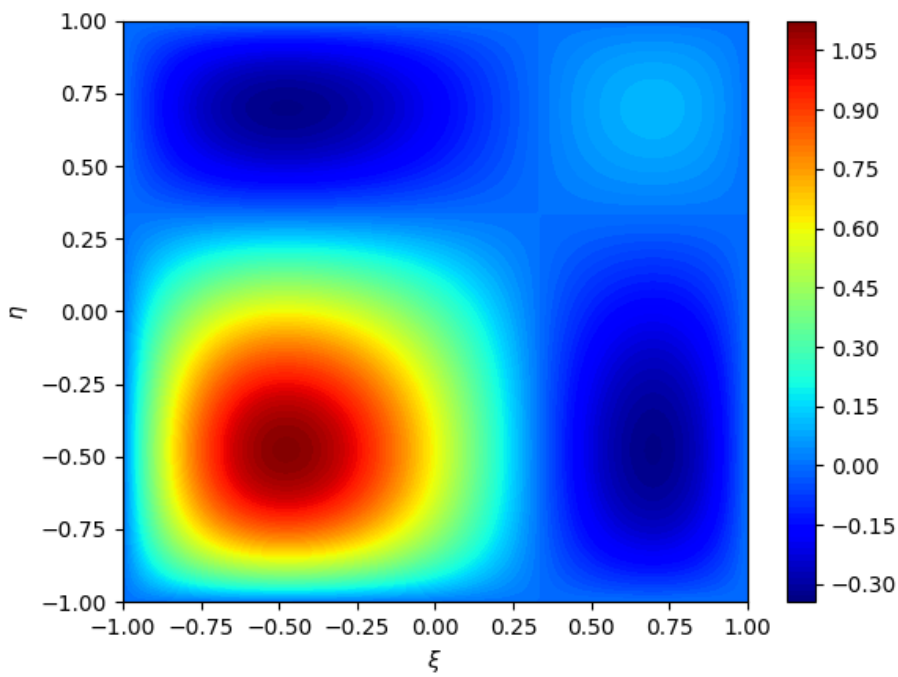
−0.50
−0.75
−1.00
η
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
ξ
1.05
0.90
0.75
0.60
0.45
0.30
0.15
0.00
−0.15
−0.30

A=5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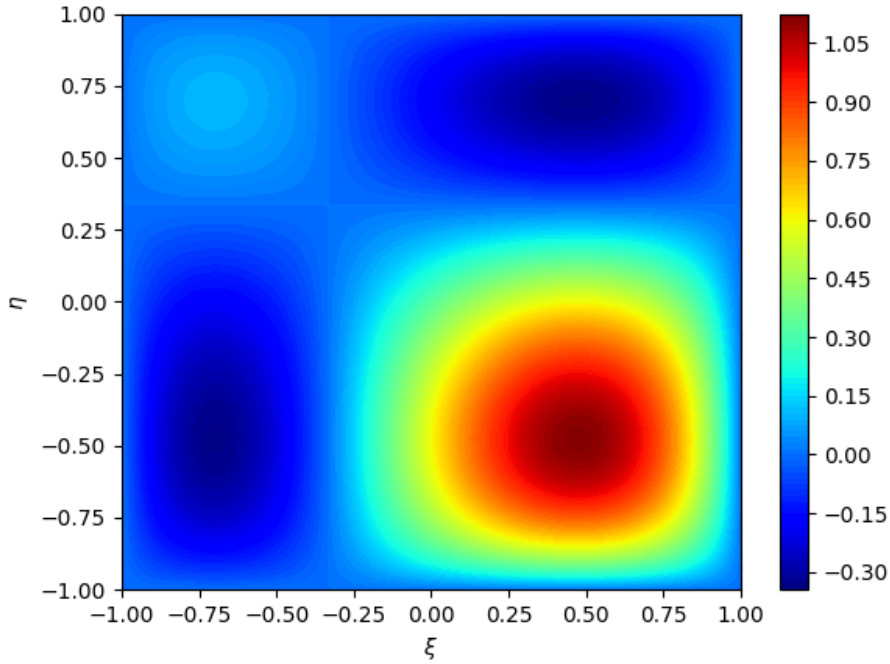
−0.50
−0.75
−1.00
η
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
ξ
1.05
0.90
0.75
0.60
0.45
0.30
0.15
0.00
−0.15
−0.30

A=6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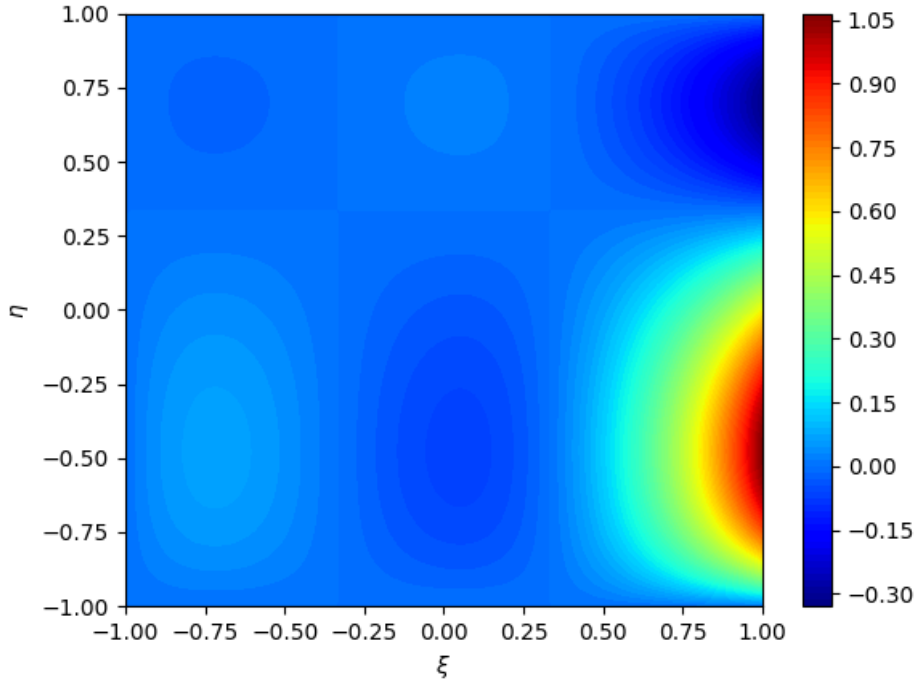
−0.50
−0.75
−1.00
η
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
ξ
1.05
0.90
0.75
0.60
0.45
0.30
0.15
0.00
−0.15
−0.30

A=7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0.50
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−0.25
−0.50
−0.75
−1.00
η
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
ξ
1.05
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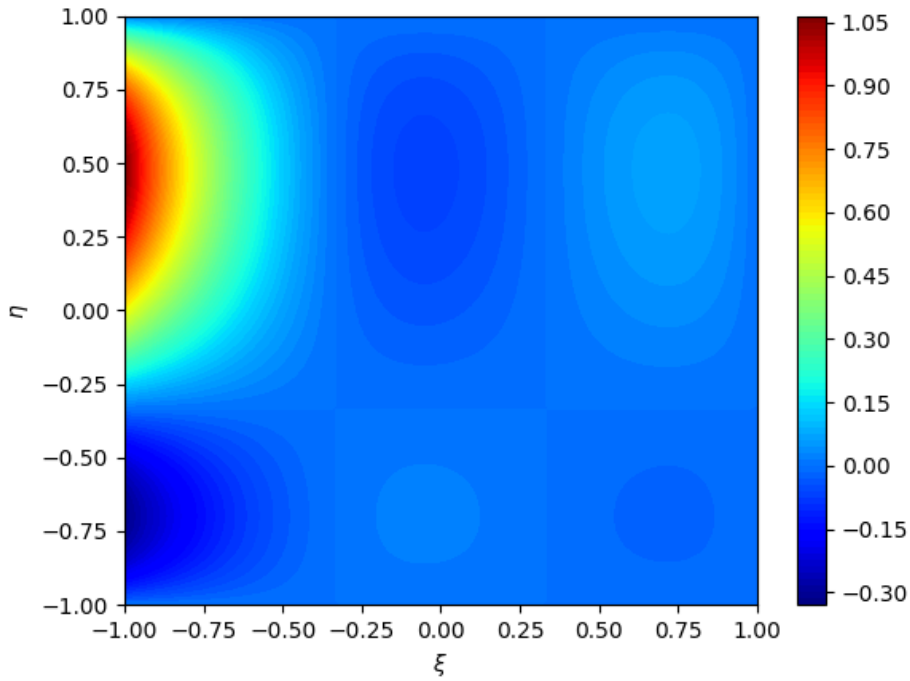
0.75
0.60
0.45
0.30
0.15
0.00
−0.15
−0.30

A=8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0.25
0.00
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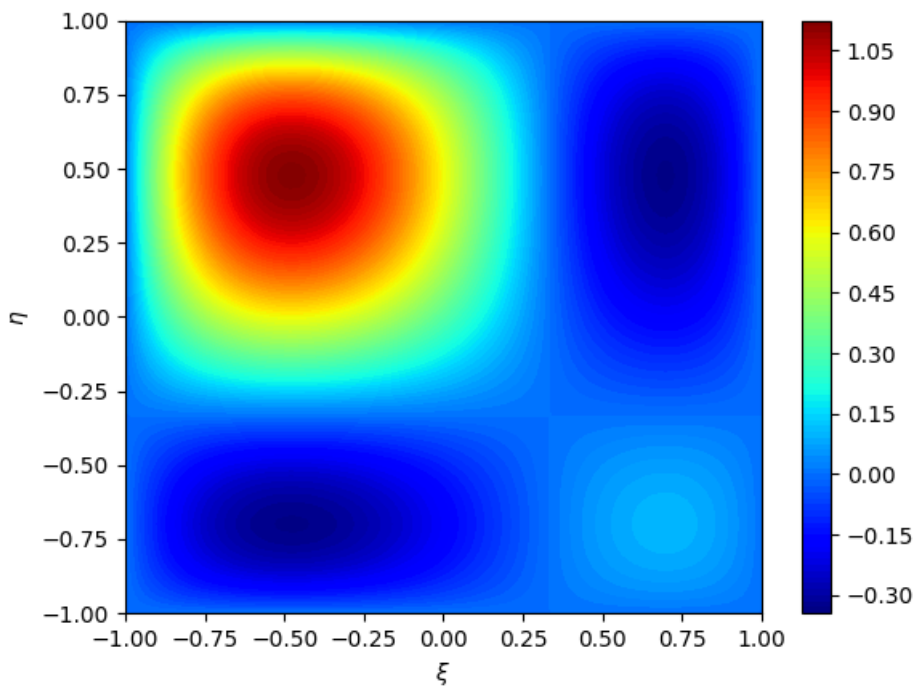
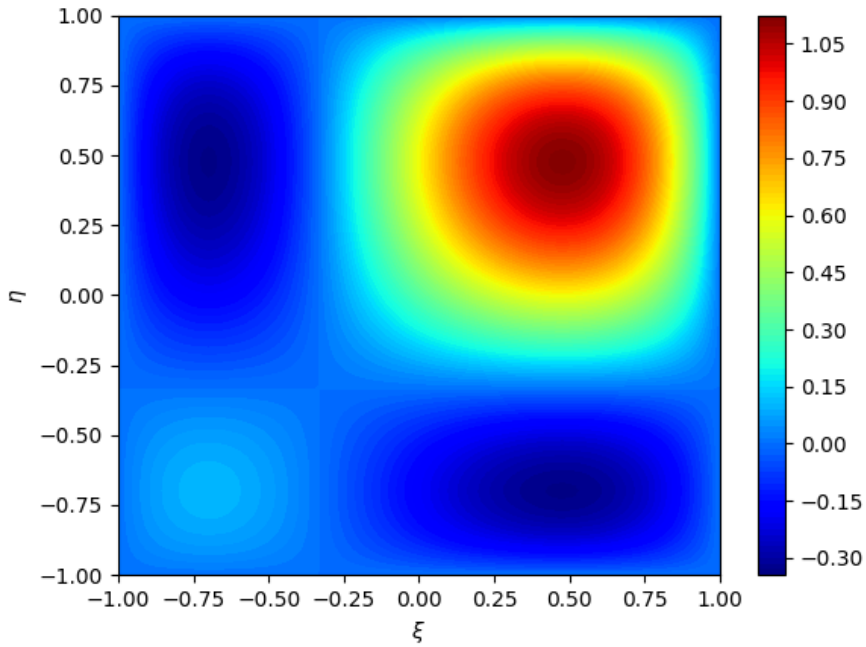
−0.50
−0.75
−1.00
η
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
ξ
1.05
0.90
0.75
0.60
0.45
0.30
0.15
0.00
−0.15
−0.30

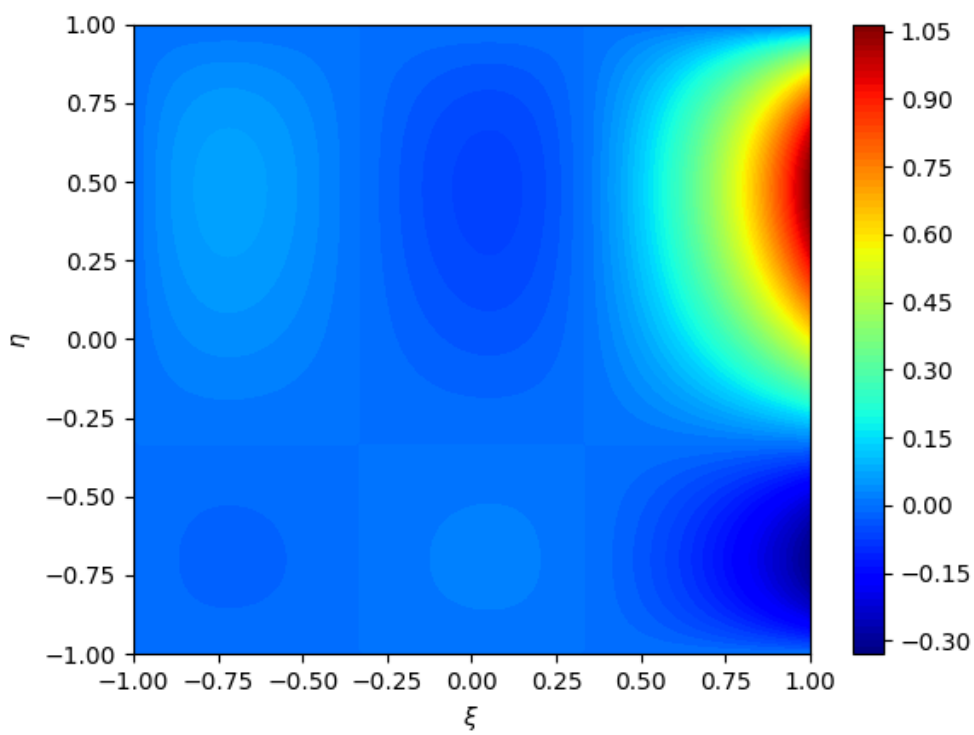
A=9


1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
η
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
ξ
1.05
0.90
0.75
0.60
0.45
0.30
0.15
0.00
−0.15
−0.30

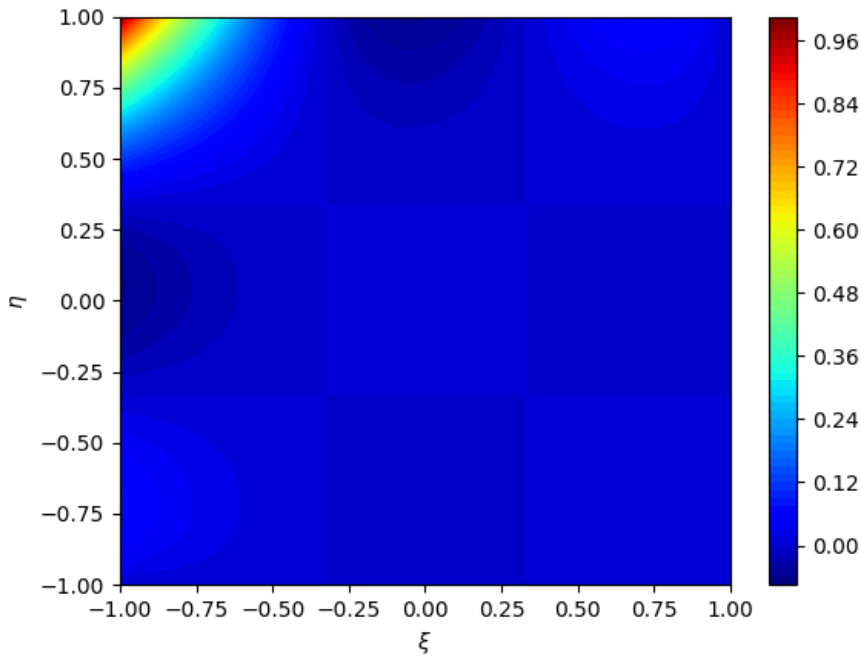
A=10


1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
η
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
ξ
1.05
0.90
0.75
0.60
0.45
0.30
0.15
0.00
−0.15
−0.30

A=11


1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
η
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
ξ
1.05
0.90
0.75
0.60
0.45
0.30
0.15
0.00
−0.15
−0.30

A=12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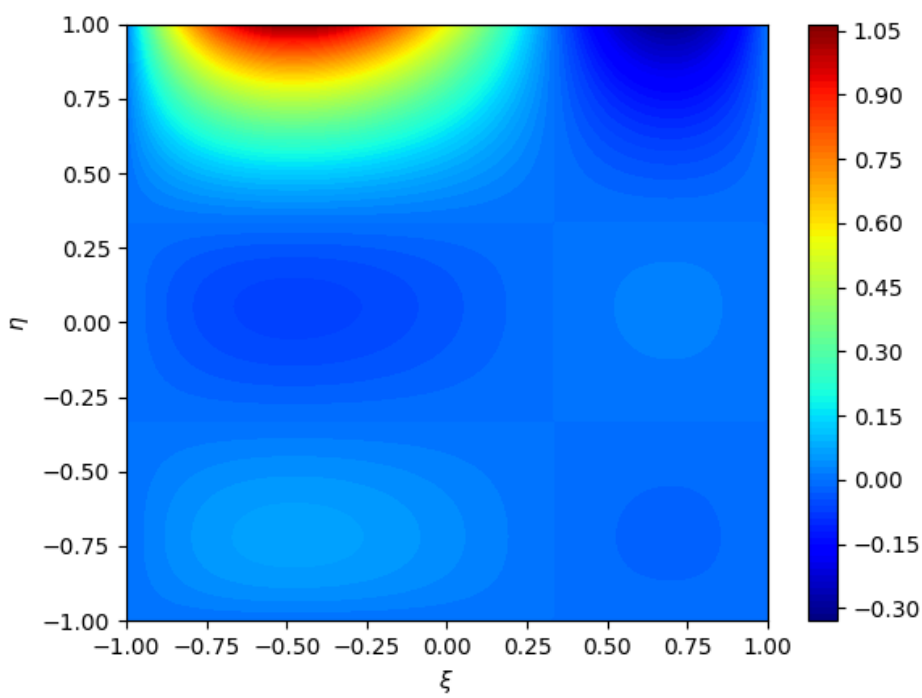
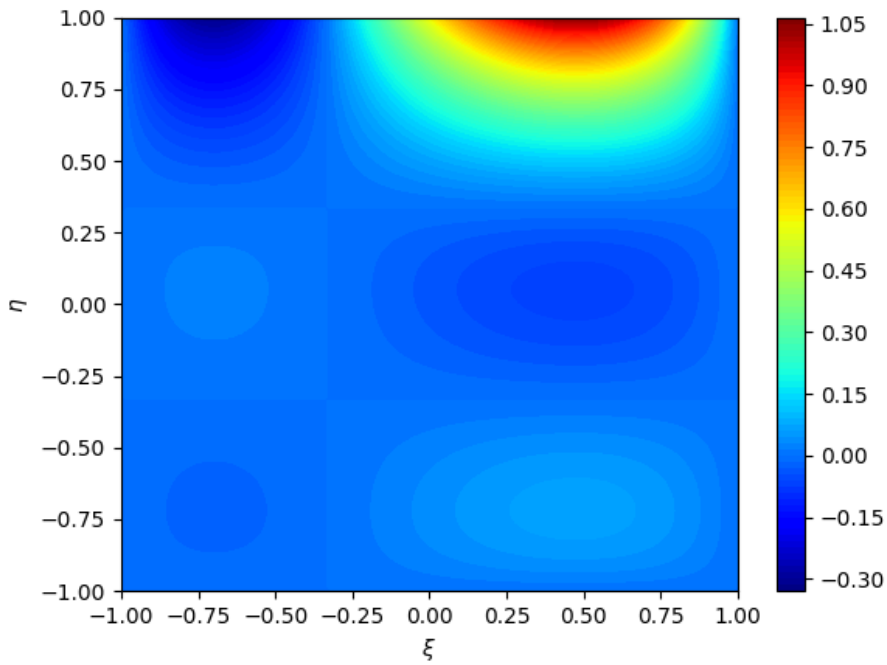
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
η
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
ξ
0.96
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00

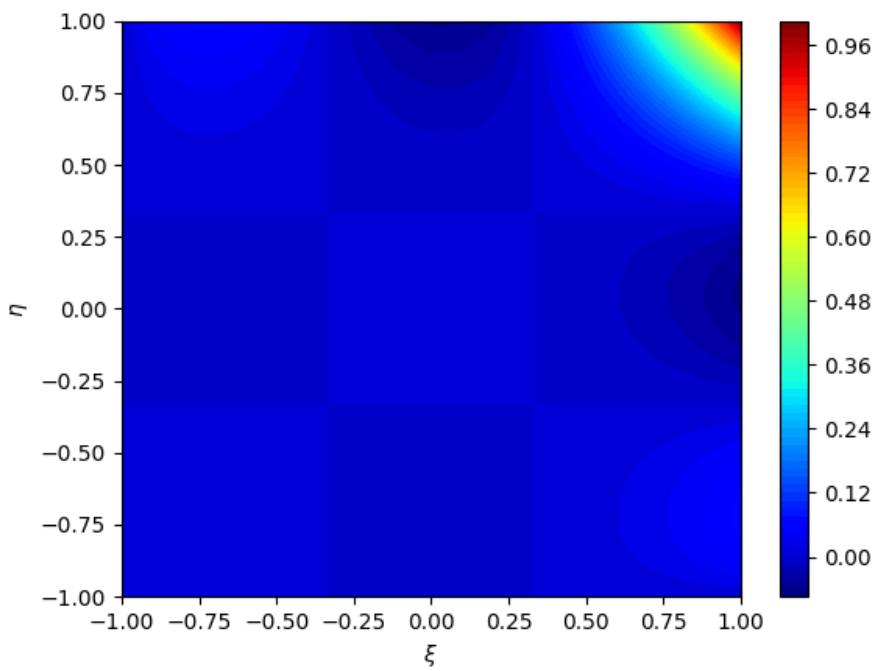
A=13


1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
η
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
ξ
1.05
0.90
0.75
0.60
0.45
0.30
0.15
0.00
−0.15
−0.30

A=14

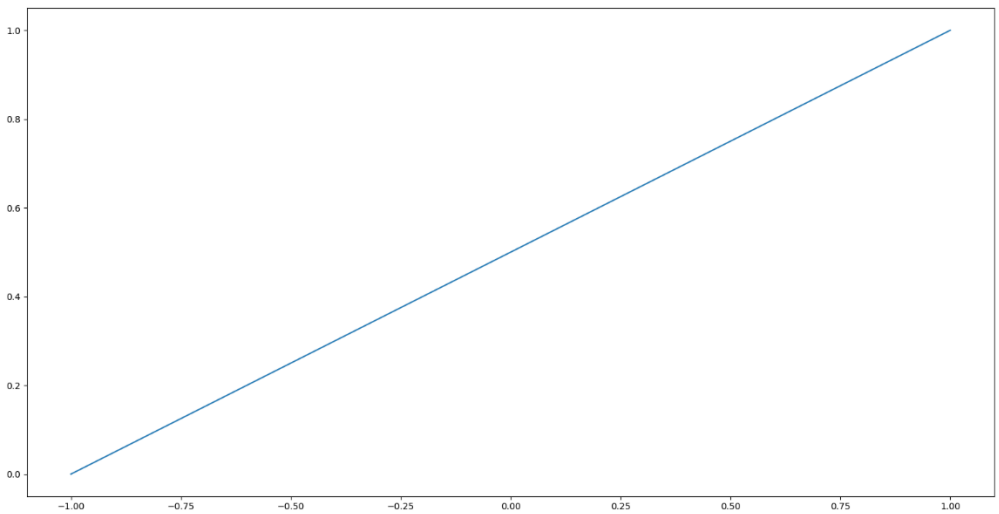

1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
η
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
ξ
1.05
0.90
0.75
0.60
0.45
0.30
0.15
0.00
−0.15
−0.30

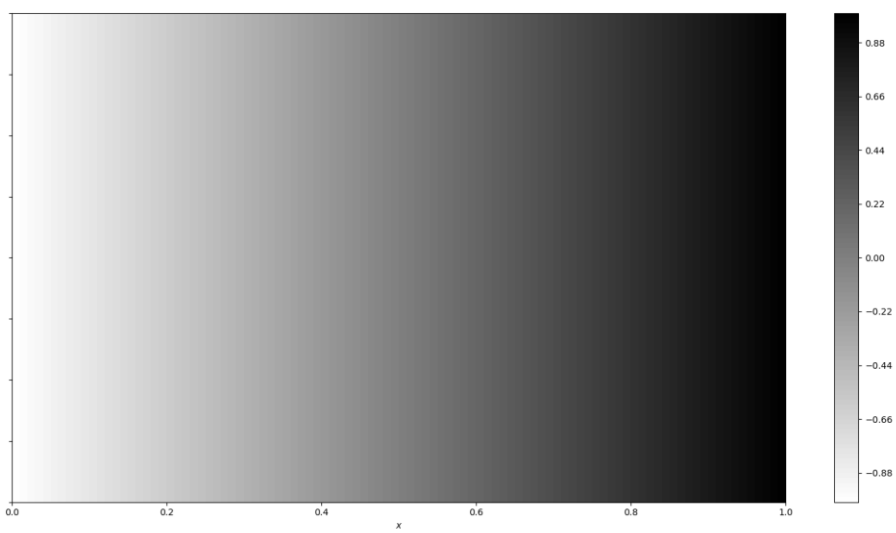
A=15


1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
η
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
ξ
0.96
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00

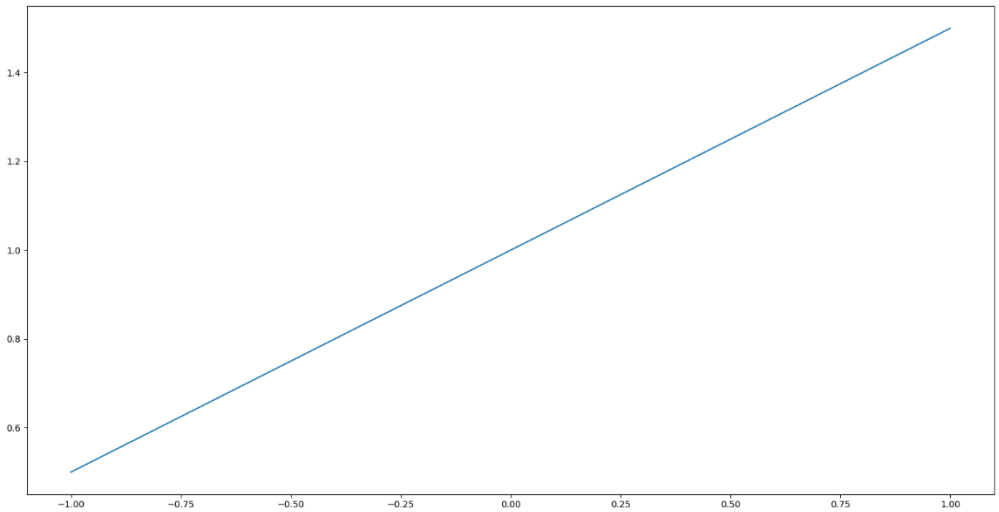
Problem 2

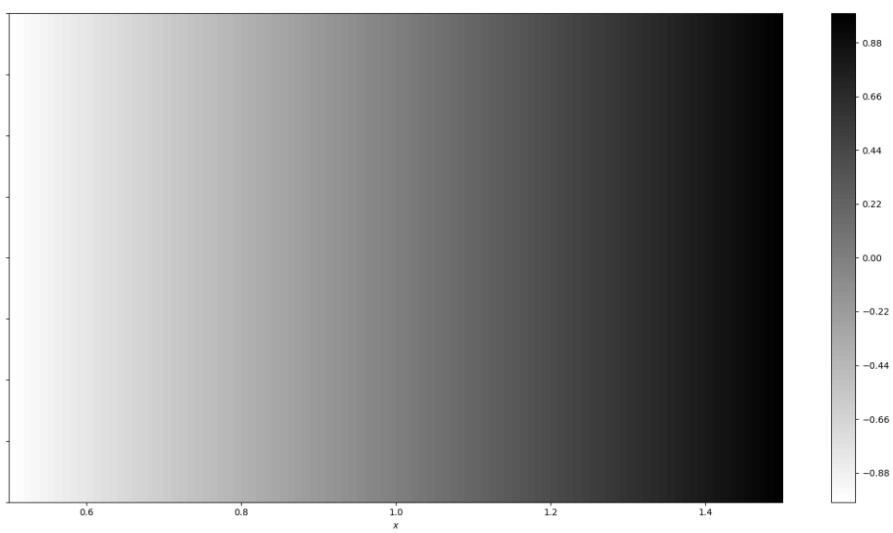
p = 1, {xa} = {0, 1}, {ξa} = {−1, 1}



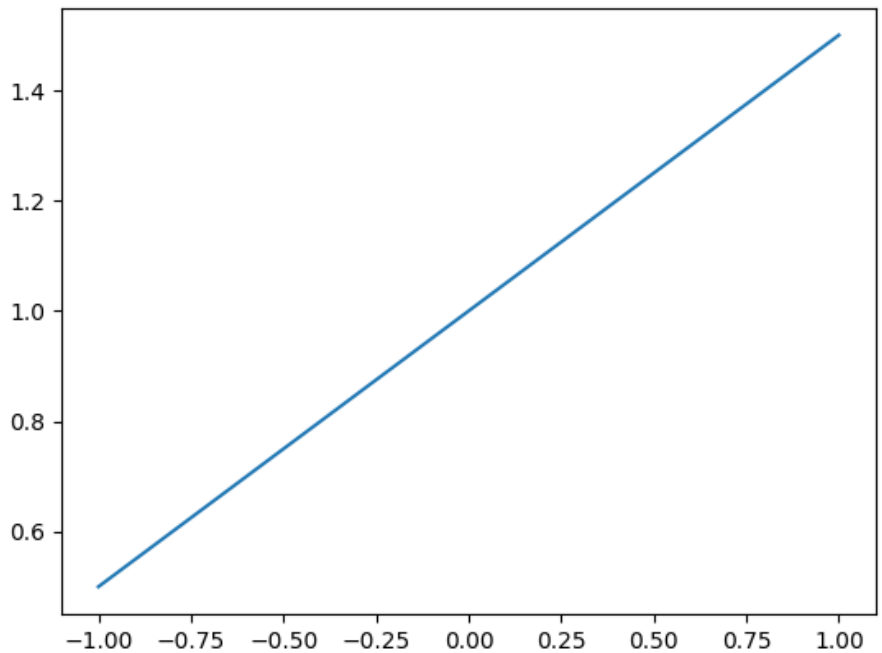


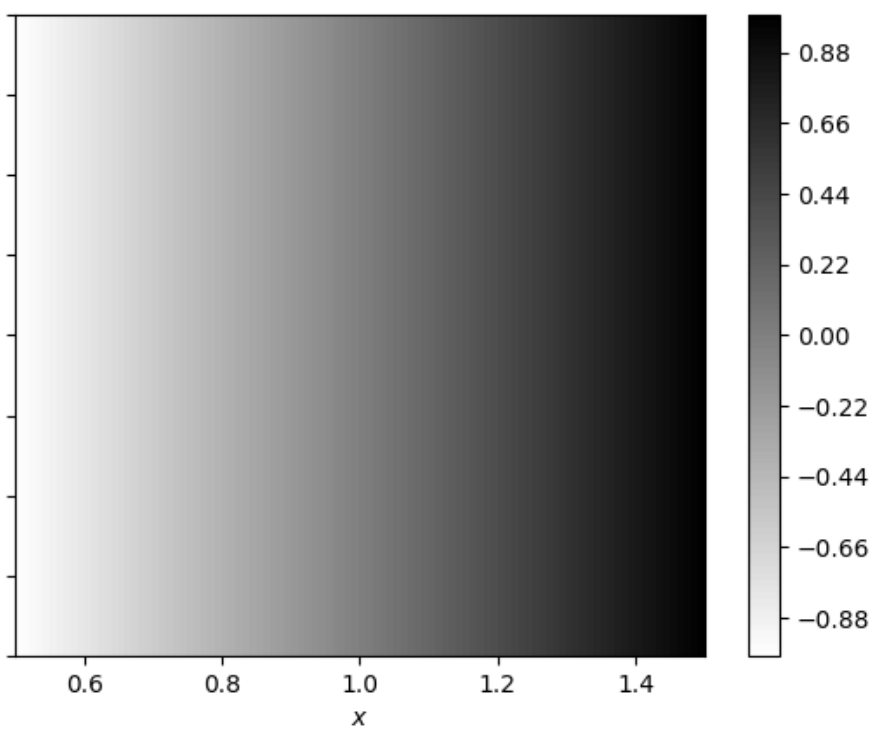
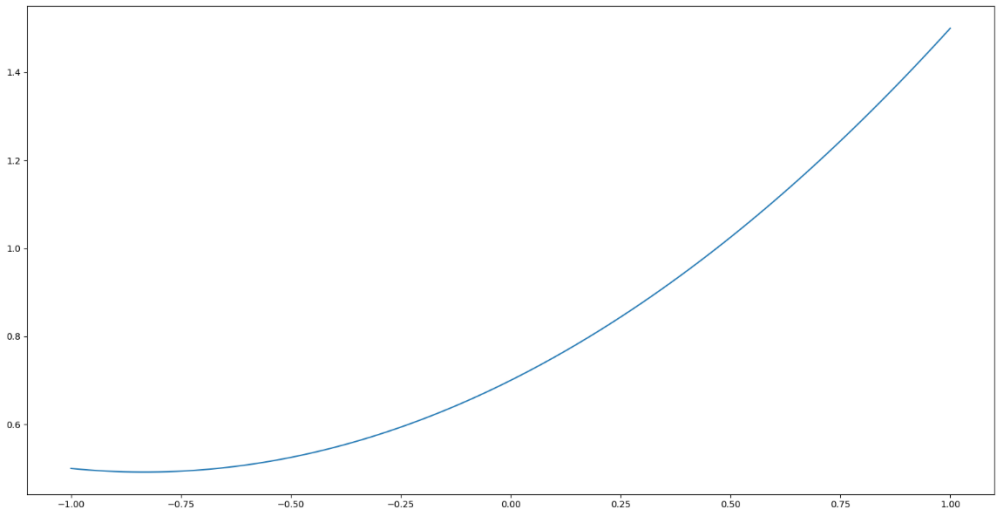
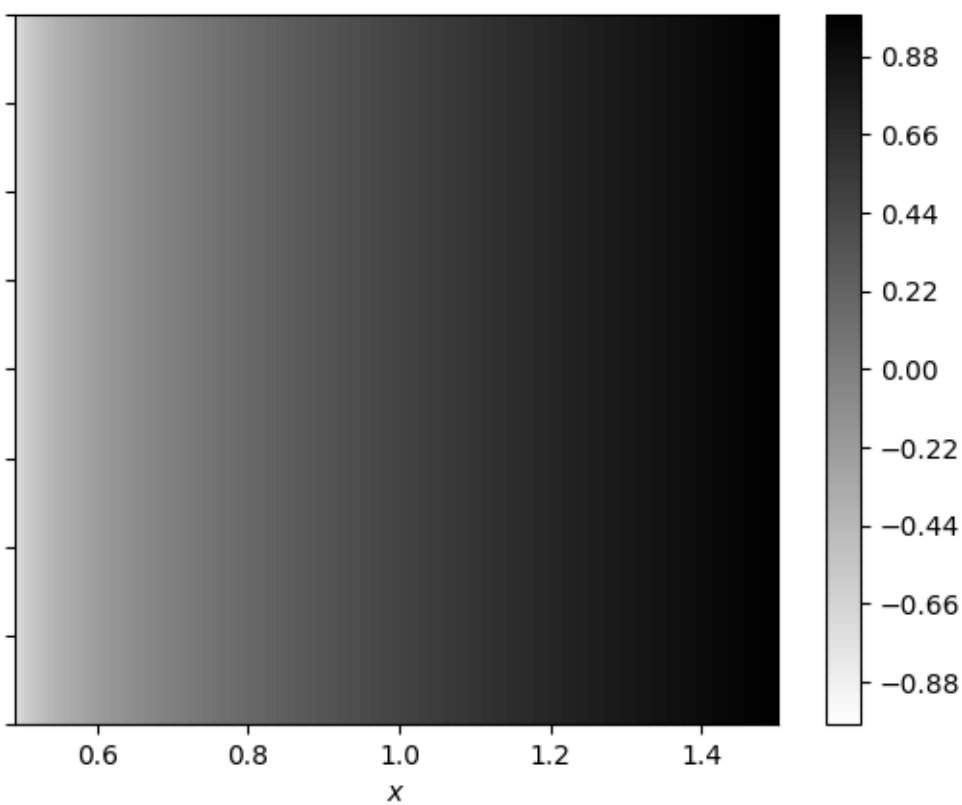
p = 2, {xa} = {0.5, 1, 1.5}, {ξa} = {−1, 0, 1}





p = 2, {xa} = {0.5, 0.7, 1.5}, {ξa} = {−1, −0.6, 1}





p = 2, {xa} = {0.5, 0.7, 1.5}, {ξa} = {−1, 0, 1}